

Statement of intent

Baha Khammari · PhD position, auction-based support schemes for renewable energy
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I finish an M.Sc. in Economics at the University of Cologne this month, with a specialisation in energy economics and a thesis in applied microeconometrics, and I am applying for the doctoral position on auction-based support schemes for renewable energy. I want to spend four years on how the design of a support scheme changes what actually gets built, and I want to do it at CORE.

CORE describes itself as an interdisciplinary centre that brings operations research, economics and econometrics together, and it names the training of doctoral researchers as one of its objectives. That is the reason I am writing to you and not to a department that does one of those three things. My own work sits on the econometrics side. What I do not yet have is graduate industrial organization and game theory at the level this project runs on, and a centre founded in 1966 on that tradition, with more than twenty professors and thirty-five doctoral students in one building, is where I would learn it properly instead of reading around it. Everything I have written so far I wrote alone, and four years inside a group that argues these questions formally is the part I want most. I should say why I want it rather than only that I do. My final seminar paper turned into a complete research design on carbon leakage from the EU ETS, with the treatment definition, the data and the identifying assumptions set against the threats to each, because I had meant to write my thesis on it. I chose a different thesis and finished the design anyway. I have been working through real analysis in my own time this year for the same reason, which is that I like the work when nobody is grading it.

What I would bring is the empirical half of your question. My thesis, graded 1.0 and worth 30 of the degree's 120 ECTS, asked whether a national scholarship programme in Brazil raised formal-sector wages and for whom. I built the panel from administrative records, five million linked employer-employee observations across 558 microregions from 2005 to 2019, in R through BigQuery, and estimated Callaway and Sant'Anna (2021) group-time effects with doubly-robust inference across exposure bins. Low-exposure regions show a wage gain of 3.04 per cent, distinguishable from zero at the 10 per cent level, on a non-monotonic dose-response that two-way fixed effects reports with the opposite sign. That skill transfers here directly. Since the 2016 state aid framework member states have adopted technology-neutral, technology-specific and mixed auction designs at different dates, which is staggered adoption of a design rather than of a policy, and the AURES II database records the design and outcome of every renewable auction in the EU and UK from 2011 to 2021. I would estimate how joint bidding shifts the awarded technology portfolio and the rent per megawatt hour against not-yet-treated member states. On the theory side my training is advanced microeconomics on strategic thinking and competition with Axel Ockenfels, graded 1.3, whose own field is market and auction design, with 1.0 in mathematics and 1.3 in econometrics behind it, and I have implemented numerical solutions to economic models. I am free from October 2026, which I know matters for a post starting this autumn, and my French is B1.

Thank you for considering my application. Prof. Dr. Pia Pinger, who chaired my thesis, and Jun.-Prof. Dr. Oliver Ruhnau, who taught and examined my module on energy markets and regulation, have both agreed to act as referees, and their contact details are on my CV. I would be glad to talk the project through whenever it suits you.

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